

LectureAssist: An Agentic Multimodal AI Assistant that Turns Lectures into Adaptive Study Material

Fully local question generation from lecture video and course documents, with no paid API anywhere in the pipeline

ABSTRACT

Students record their lectures but rarely get the practice questions that would make those recordings useful. LectureAssist closes that gap. It ingests a raw lecture video or a course document, classifies every sampled frame as a slide, a blackboard, a whiteboard or a plain scene, and preprocesses each class differently before reading it with OCR. Whisper transcribes the audio in parallel. The two channels fuse into one time-stamped timeline that is summarised, embedded and indexed. A team of agents then plans topics, drafts questions, checks each one against the lecture and rewrites whatever fails. Nothing leaves the machine.

Method with System Diagram/Design Complexity

An orchestrator drives a team of specialist agents. Frames are sampled and audio demuxed; documents are parsed directly. Brightness and edge statistics route each frame to its own contrast enhancement, thresholding and polarity inversion before EasyOCR reads it, while Whisper handles speech. A planner allocates topic and difficulty slots. A controller then assigns every slot one cognitive track, so recall, inference and synthesis questions are enforced rather than hoped for. The generator drafts one question at a time and may query the FAISS index itself when its context looks thin. A validator diagnoses failures and a refiner acts on it.

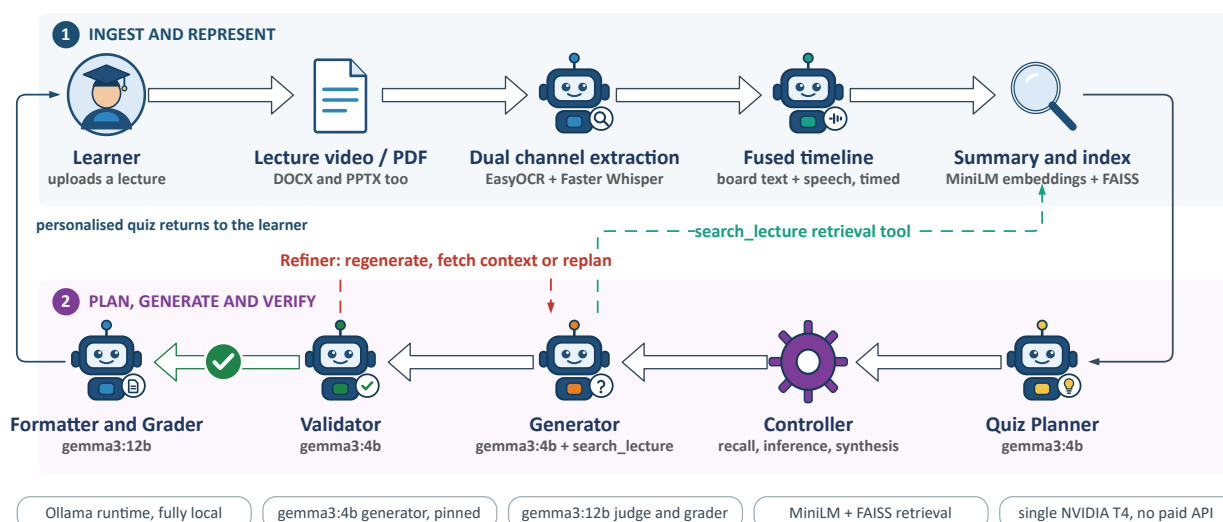


Figure 1: LectureAssist architecture, drawn as the loop it is. Stage 1 turns raw media into one grounded representation. Stage 2 plans, generates and verifies: a failed question is diagnosed and returned to the Refiner rather than discarded, and the formatted quiz goes back to the learner it started from.

Novelty of Project

Two things are genuinely new. Extraction adapts to what it sees instead of applying one OCR recipe, so chalk on a dark board survives alongside projected slides. And the validator diagnoses a failure, then picks the fix: rewrite, fetch more lecture context, or abandon the topic. Comparable tools only filter.

Impact on Society/Environment

Self testing is the strongest way to study, yet practice questions stay a privilege of rich institutions. LectureAssist generates them from a student's own lectures on hardware they already own, removing a cost barrier, not an ability one. A four billion parameter model on one GPU also spares a datacentre.

Business Model/Feasibility/Financial Scalability Plan

There is no API bill, so the marginal cost of one more student is essentially zero. That shapes the plan. The local

build stays free and open source for individuals, reaching students who cannot pay at all. Departments buy the tier that matters to them: one GPU box serving a whole course, licensed per seat per semester at a fraction of cloud pricing, with LMS integration and support as the paid work. Feasibility is not theoretical. The entire system already runs inside a free Colab session on a single T4, so one modest GPU can serve a class.

Conclusion

Every stage of LectureAssist is built and running: ingestion, scene aware extraction, retrieval, question generation, validation, grading and adaptive difficulty, all on one consumer GPU. What remains is breadth rather than depth. Longer multi hour recordings, Bengali lecture audio, and a classroom trial with real students are the next milestones.